

Asian Transport Database Overview

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1 Literature review

Completed a literature review to understand datasets/ complete work already published and open source.

Outputs:

Asia Transport Literature Tracker.xlsx

- Includes relevant publications, reports, datasets, organisations, list of countries in study area and modes relevant to each country.

Asia Transport Literature Tracker.pdf

- Summary of spreadsheet

2 Main Spatial File Outputs

3 Airports

3.1 Data sources used

3.2 Analysis done

Airport node spatial data correction (completed in Airport.py)

The World Bank airport dataset provided the only identified open source data containing flight connection paths between airports. Therefore, the World Bank airport points were used as the basis for the airport nodes used for the flight network.

However, the World Bank airport coordinates were found to have poor spatial accuracy. The airport locations were therefore checked and corrected using the OurAirports dataset, which appeared to provide better spatial accuracy.

The analysis used airport data from Ourairports.csv, worldbank_filtered_airport_flows.gpkg and worldbank_filtered_airport_volume.gpkg.

OurAirports data was first filtered to include only countries within the study area using Countries_list.xlsx. The World Bank airport coordinates were then corrected using the OurAirports coordinates where a matching IATA code was available. Where no matching OurAirports record was available, the original World Bank coordinates were retained.

The origin and destination coordinates of the World Bank flight connections were then corrected using the corrected airport coordinates.

airport_coordinate_audit.xlsx contains the audit of this matching process. The match_summary sheet provides a summary of the number of airport locations that were common between the two datasets. 14 airports in the World Bank dataset could not be matched and therefore could not be spatially corrected using OurAirports.

3.2.1 Shapefile matching (OSM_extract_country_loop and airport_shapefile_nearest.py):

Relevant tags were extracted from OSM.

For Aiports these were:

```
{"aeroway": ["aerodrome"]},  
{"aeroway": ["terminal"]},
```

```
{"aeroway": ["runway"]},
```

```
{"aeroway": ["taxiway"]},
```

From here, any extracted OSM features located within 2 km of an airport node in `world-bank_filtered_airport_volume.gpkg` were assigned to that airport. The airport's IATA code was then added to the feature's attributes. The final output combines all extracted OSM airport features into a single layer, with attributes identifying the feature's original OSM tag (e.g. aerodrome, terminal, runway or taxiway) and the airport it was assigned to.

3.2.2 Still to do:

- Check the spatial location of the 14 airport nodes in `airport_flows_world_bank_corrected.gpkg`, which did not have a match to the OurAirports database. (Validating that the flows also update if any airports get updated)
- Add missing polygon data to the XX airports

4 Maritime

4.1 Data sources used/outputted

- Port watch
- Multi-hazard risk to global port infrastructure and resulting trade and logistics losses, [Jasper Verschuur](<https://www-nature-com.ezproxy-prd.bodleian.ox.ac.uk/articles/s43247-022-00656-7#auth-Jasper-Verschuur-Aff1>)
- Port_landuse.gpkg (shapefiles for ports)
- Global_maritime_network.gpkg (each node is at the centre of the port_landuse shapefiles that it is relevant to)
- Ports.shp

4.2 Analysis done

4.2.1 Maritime network (completed in ports.py):

Code initially copied from the `African-transport-dataset/scripts/preprocess/ "ports_new_merge.py"`. Using the `Ports.shp` and updated 2023-2025 attribute data provided by Jasper Verschuur (`port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`). A node and edge network was created, with the attributes. The network was broken up into 3 clusters, to aid visuals (`northern_russia`, `pacific_network` and `large_cluster`). The output contains two node types, ports (land) and maritime (networking), alongside the edge connections between.

-file output names

`global_maritime_network_PROVA_NEW1`

`asia & pacific_maritime_network_PROVA_NEW1`

`northern_russia_network_maritime_network`

`large_cluster_maritime_network`

`pacific_network_maritime_network`

4.2.2 Shapefile correction (all done using `port_shapefile_nearest.py` and manually):

`port_landuse.gpkg` shapefiles were matched to their corresponding ports by assigning the nearest port node to each polygon (nodes stored in `asia_pacific_maritime_network_PROVA_NEW1.gpkg`). The results of this automated matching are stored in `port_landuse_nearest_port_id_lookup.xlsx`.

The accuracy of these matches was then manually checked and recorded in `port_landuse_nearest_port_id_lookup.xlsx`. Each match was assessed using the distance between the port node and the land-use polygon, the names associated with both datasets, and the port's location on Google Maps. In the checked file, the checked column indicates whether the port land-use shapefile (Column A) has been matched to the correct port node (Column H). Where checked indicates that the match is incorrect, the correct port id column records the correct port ID, identified manually.

The corrections were then applied, and the `port_landuse_nearest_port_ids.gpkg` and `port_landuse_nearest_port_ids.shp` contain the correct results.

14 ports from the `asia_pacific_maritime_network_PROVA_NEW1.gpkg` file did not have any polygon data (from `port_landuse.gpkg`) matched to it. Seen in `port_landuse_nearest_port_id_lookup_checked.xlsx` sheet-node_polygon_audit, column `missing_polygon_data = TRUE`.

4.2.3 Still to do:

- Add missing polygon data to the 14 ports.

5 Road

Primary roads were extracted from OSM using the OpenGIRA tool. The OpenGIRA tool was run on ARC, and the outputs were then combined into a single layer using `Combine_opengira.py`.

The OSM data for 5 regions was used to cover the entire study area. They were Asia, Oceania, Russia, Turkey and Cyprus. (pbf coverage can be seen at <https://download.geofabrik.de/>)

The data was extracted from opengira using the filter `road-primary`. The the 5 regions spatially overlapped, hence `Combine_opengira.py` was used to combine the outputs into a single layer, removing any duplicated edges and nodes.

For road edges, the duplications were removed based on the `osm_way_id` attribute, and for road nodes, the duplications were removed based on spatial location, due to there not being an `osm_node_id` for primary road nodes.

The final output is stored in `asia_pacific_road_primary_edges.gpq` and `asia_pacific_road_primary_nodes.gpq`.

6 Rail

Rail lines were extracted from OSM using the OpenGIRA tool. The OpenGIRA tool was run on ARC, and the outputs were then combined into a single layer using `Combine_opengira.py`.

The OSM data for 5 regions was used to cover the entire study area. They were Asia, Oceania, Russia, Turkey and Cyprus. (pbf coverage can be seen at <https://download.geofabrik.de/>)

The data was extracted from opengira using the filter `rail`. The the 5 regions spatially overlapped, hence `Combine_opengira.py` was used to combine the outputs into a single layer, removing any duplicated edges and nodes.

For both rail edges and nodes, the duplications were removed based on the `osm_way_id` and `osm_node_id` respectively attribute. The final output is stored in `asia_pacific_road_primary_edges.gpq` and `asia_pacific_road_primary_nodes.gpq`.

and asia_pacific_road_primary_nodes.gpq.

7 Inland waterways

Investigated OSM tags, appearing to best map inland waterways:

Tag = boat:yes

Tag = ferry : terminal

8 Multimodal

9 ARC

ARC is separated into two file directories:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/asia-transport-database
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

10 Using open-gira on ARC

Purpose: Used to create the road and rail networks using OpenGIRA.

1. Go to the OpenGIRA directory:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

2. Activate the OpenGIRA environment:

```
module load Mamba/23.11.0-0
export CONDA_ENVS_PATH=/home/mans4968/.mamba/envs
export MAMBA_ROOT_PREFIX=/home/mans4968/.mamba
eval "$(conda shell.bash hook)"
conda activate /home/mans4968/.mamba/envs/open-gira
```

3. Run OpenGIRA using Snakemake and the ARC profile (not run on a .sh file):

```
snakemake -profile config/ARC -cores 1
```

A specific output can also be requested, for example:

```
snakemake -profile config/ARC -cores 1 results/russia-latest_filter-road-primary/edges.
```

4. Check jobs currently running:

```
squeue -u $USER
```

5. Check completed/failed jobs:

```
sacct -u $USER -format=JobID,JobName,State,Elapsed,ExitCode
```

6. Outputs are stored in:

```
/data/cenv-opsis-astdb/\{}<USERNAME>/open-gira/results/
```

Completed final networks are organised into:

```
results/road/
```

```
results/rail/
```

Other folders such as geoparquet/, slices/, json/ and input/ contain input or intermediate OpenGIRA data.

11 Asia-transport-database on ARC

Purpose: Used to run the Asia transport database processing, including the country-by-country OSM infrastructure extraction.

1. Go to the Asia database:

```
cd /data/cenv-opsis-astdb/\{}<USERNAME>/asia-transport-database
```

2. Input OSM data are stored in:

```
incoming_data/osm/
```

3. Run the required Asia transport database script/job on ARC.

See various .sh files

4. Processed outputs are stored in:

```
processed_data/infrastructure/
```

12 Mistral

13 Already existing country multimodal transport packages

Multimodal transport data packages have already been extracted for China, Singapore, New Zealand, and Australia. These datasets can be used later for comparison and validation against the wider transport database. The packages are stored under:

```
incoming_data/infrastructure/[folder name]
```

The content of each country package varies, and not all packages contain data for every transport mode.

““latex

14 Files Overview (incl. Directories, descriptions, references, and processing scripts)

Catalogue of data inputs, intermediate files, final outputs, and processing scripts.

Files are grouped by repository location and document their purpose, source, and associated processing script.

-
1. `airport_flows_world_bank_corrected.gpkg`
Directory: `processed_data/infrastructure/airport/`
Description: World Bank airport flow data with origin and destination coordinates corrected using the corrected airport locations.
Reference: World Bank and OurAirports.
Produced by: `Airports.py`

 2. `airport_osm_features_within_2km.gpkg`
Directory: `processed_data/infrastructure/airport/`
Description: OSM airport lines and polygons located within 2 km of corrected World Bank airport points and assigned to the relevant airport.
Reference: OpenStreetMap and corrected World Bank airport data.
Produced by: `airport_shapefile_nearest.py`

 3. `airport_osm_features_within_2km.xlsx`
Directory: `processed_data/infrastructure/airport/`
Description: Audit workbook containing matched OSM features, airport summaries, and OSM tag summaries.
Reference: OpenStreetMap and corrected World Bank airport data.
Produced by: `airport_shapefile_nearest.py`

 4. `airport_coordinate_audit.xlsx`
Directory: `processed_data/infrastructure/airport/`
Description: Audit spreadsheet showing which World Bank airports were matched to OurAirports and spatially corrected, and identifying unmatched World Bank airports.
Reference: World Bank and OurAirports.
Produced by: `Airports.py`

 5. `asia_pacific_road_primary_edges.gpq`
Directory: `processed_data/open_gira/results/road_primary/combined/`
Description: Primary road edge in the Asia-Pacific study area, removing duplicates from different sources.
Reference: <https://github.com/nismod/open-gira>
Produced by: `OpenGira` and `Combine_opengira.py`

 6. `asia_pacific_road_primary_nodes.gpq`
Directory: `processed_data/open_gira/results/road_primary/combined/`
Description: Primary road nodes in the Asia-Pacific study area, removing duplicates from different sources.
Reference: <https://github.com/nismod/open-gira>
Produced by: `OpenGira` and `Combine_opengira.py`
-

7. [country/region]_[tag].parquet

Directory: processed_data/infrastructure/osm_filter/

Description: OSM features extracted for specified tags from country or regional PBF files, including individual-country and wider study-area outputs where specified.

Reference: Geofabrik OpenStreetMap extracts.

Produced by: OSM_extract_country_loop.py

8. [country/region].osm.pbf

Directory: incoming_data/osm/

Description: Raw country or regional OpenStreetMap PBF extracts used for OSM feature extraction.

Reference: Geofabrik OpenStreetMap extracts.

Source/Produced by: Geofabrik / OpenStreetMap.

9. Countries_list.xlsx

Directory: incoming_data/

Description: List of all countries included in the study area.

Reference: Created by Darcey Sutcliffe.

10. filtered_Ourairports.csv

Directory: processed_data/infrastructure/airport/

Description: OurAirports data filtered to include small, medium, and large airports within the study area.

Reference: OurAirports.

Produced by: Airports.py

11. filtered_Ourairports.gpkg

Directory: processed_data/infrastructure/airport/

Description: Spatial version of the filtered OurAirports data containing small, medium, and large airports within the study area.

Reference: OurAirports.

Produced by: Airports.py

12. global_maritime_network_PROVA_NEW1.gpkg

Directory: processed_data/infrastructure/port/

Description: Global maritime network containing port and maritime network nodes and their connections.

Reference: Maritime port data and PortWatch attributes.

Produced by: ports.py

13. `cleaned_asia_pacific_maritime_network_PROVA_NEW1.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Final cleaned Asia-Pacific maritime network combining the three separate network clusters.
Reference: Northern Russia, large cluster, and Pacific maritime networks.
Produced by: `ports.py`

14. `asia_pacific_maritime_network_PROVA_NEW1.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Asia-Pacific maritime network covering the study area and including connections between the separate network clusters.
Reference: Maritime port data and PortWatch attributes.
Produced by: `ports.py`

15. `large_cluster_maritime_network.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Maritime network representing the large cluster containing the majority of Asian network nodes.
Reference: Maritime port data and PortWatch attributes.
Produced by: `ports.py`

16. `northern_russia_network_maritime_network.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Maritime network representing the northern Russia cluster.
Reference: Maritime port data and PortWatch attributes.
Produced by: `ports.py`

17. `pacific_network_maritime_network.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Maritime network representing the Pacific cluster.
Reference: Maritime port data and PortWatch attributes.
Produced by: `ports.py`

18. `port_calls_average_2023-2025.csv`
Directory: `incoming_data/Global port supply-chains/`
Description: Port attribute data containing average port calls for 2023–2025.
Reference: PortWatch, provided by Jasper Verschuur. PortWatch.

19. `port_capacity_called_average_2023-2025.csv`
Directory: `incoming_data/Global port supply-chains/`
Description: Port attribute data containing average port capacity called for 2023–2025.
Reference: PortWatch, provided by Jasper Verschuur. PortWatch.

20. `port_landuse.gpkg`
Directory: `incoming_data/infrastructure/port/`
Description: Polygon data representing maritime port land use.
Reference: Mendeley Data.
Source: Jasper Verschuur.
-
21. `port_landuse_nearest_port_id_lookup.xlsx`
Directory: `processed_data/infrastructure/port/`
Description: Lookup showing the nearest port assigned to each port land-use polygon before manual corrections.
Reference: Port land-use data and the maritime network.
Produced by: `port_shapefile_nearest.py`
-
22. `port_landuse_nearest_port_id_lookup_checked.xlsx`
Directory: `incoming_data/infrastructure/port/`
Description: Manually checked lookup assessing each nearest-port assignment and recording the correct `port_id` where required.
Reference: Created by Darcey Sutcliffe.
Source/Produced by: Manual validation by Darcey Sutcliffe.
-
23. `port_landuse_nearest_port_id_lookup_checked_result.xlsx`
Directory: `processed_data/infrastructure/port/`
Description: Validation spreadsheet comparing original and final `port_id` values to confirm that manual corrections were applied correctly.
Reference: Manually checked port lookup.
Produced by: `port_shapefile_nearest.py`
-
24. `port_landuse_nearest_port_ids.gpkg`
Directory: `processed_data/infrastructure/port/`
Description: Final GeoPackage containing the corrected relationship between each port land-use polygon and its corresponding `port_id`.
Reference: Port land-use data and the maritime network, with manual corrections applied.
Produced by: `port_shapefile_nearest.py`
-
25. `port_turn_around_time_average_2023-2025.csv`
Directory: `incoming_data/Global port supply-chains/`
Description: Port attribute data containing average port turnaround time for 2023–2025.
Reference: PortWatch, provided by Jasper Verschuur. PortWatch.
-

26. `Ports.csv`
Directory: `incoming_data/infrastructure/port/`
Description: Point locations of maritime ports.
Reference: Mendeley Data.
Source: Jasper Verschuur.
-
27. `Ourairports.csv`
Directory: `incoming_data/infrastructure/Airport/`
Description: OurAirports airport attributes and coordinates, containing the global data reported by OurAirports.
Reference: OurAirports.
Source/Produced by: External dataset.
-
28. `utils_new.py`
Directory: `scripts/preprocess/`
Description: Utility functions used by the preprocessing scripts, copied from the African-transport-dataset project.
Reference: `African-transport-dataset/scripts/preprocess/`
Produced by: Copied and adapted from the African-transport-dataset project.
-
29. `worldbank_filtered_airport_flows.gpkg`
Directory: `incoming_data/infrastructure/Airport/`
Description: World Bank airport flow data filtered to include only flows between airports within the study area.
Reference: World Bank Global Airports dataset.
Source/Produced by: External dataset.
-
30. `worldbank_filtered_airport_volume.gpkg`
Directory: `incoming_data/infrastructure/Airport/`
Description: World Bank airport data filtered to include only countries within the study area.
Reference: World Bank Global Airports dataset.
Source/Produced by: External dataset.
-
31. `world_bank_airports_corrected.gpkg`
Directory: `processed_data/infrastructure/airport/`
Description: World Bank airport locations corrected using OurAirports coordinates where a matching IATA code was available; original World Bank coordinates are retained where no match was found.
Reference: World Bank and OurAirports.
Produced by: `Airports.py`